INCH-POUND

MIL-G-49469(HD)
17 APRIL 1991

MILITARY SPECIFICATION
GENERATOR, HAND CRANKED
G-67B/G

AMSC: N/A

FSC-6115

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1. SCOPE

1.1 This specification covers requirements for Generator, Direct Current, G-67B/G, a portable, hand-cranked electrical generator to provide power to military radios, coding equipment, rechargeable batteries, and other electronic equipment in the field.

2. APPLICABLE DOCUMENTS

2.1 Government Documents

2.1.1 Specifications, Standards, and Handbooks. The following specifications, standards, and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

Federal Standards

FED-STD-595	Colors Used in Government Procurement

Military Specifications

MIL-A-8625	Anodic Coatings, for Aluminum and Aluminum Alloys
MIL-A-63576	Aluminum Oxide Coatings, Lubricative, for Aluminum and Aluminum Alloys
MIL-P-11268	Parts, Materials and Processes Used in Electronic Communication Equipment
MIL-M-13231	Marking of Electronic Items
MIL-F-14072	Finishes for Ground Electronic Equipment

Military Standards

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-252	Wired Equipment, Classification of Visual and Mechanical Defects
MIL-STD-454	Standard General Requirements for Electronic Equipment

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MIL-HDBK-781 Reliability Design Qualification and Production Acceptance Tests: Exponential Distribution

2.1.2 Other Government Documents, Drawings, and Publications. The following other Government documents, drawings and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

Drawings

USA LABCOM Harry Diamond Laboratories (FSCM 19202)

PL A3174122 Parts List for Generator, Direct Current: G-67B/G with Bag

DL A3174122 Data List

EDL A3174122 Equipment Data List

(Copies of specifications, standards, handbooks, drawings, publications and other Government documents required by suppliers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Order of Precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Definition. The Generator, Direct Current, G_67B/G is a portable, hand-cranked electrical generator to provide power to military radios, coding equipments, rechargeable batteries, and other equipments in the field.

3.2 Characteristics.

3.2.1 Performance requirements. The performance requirements contained herein are mandatory. Tests to be performed at ambient temperature unless otherwise specified.

3.2.1.1 Preconditioning.

3.2.1.1.1 Burn-in. Prior to delivery and after all processing, each Generator Head Assembly (A3174181) shall be burned in (See 4.2.4.1).

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3.2.1.1.2 Bounce preconditioning. The Generator Head Assembly (A3174181), shall meet the performance requirements of 3.2.1.2 after being subjected to bounce preconditioning (See 4.2.4.2).

3.2.1.2 Functional Requirements.

3.2.1.2.1 Rated output (resistive load). The Generator Head Assembly (A3174181) shall provide a current of not less than 2.4 amperes into a resistive load of 10 ohms with an output voltage of not less than 24 volts at a cranking speed of 85 rpm. (See 4.2.4.3).

3.2.1.2.2 Regulated output. The regulated output voltage of the Generator Head Assembly (A3174181) while cranking at 105 rpm into a 100-ohm load shall not be less than 30 volts and shall not exceed 32.0 volts DC at ambient temperature and shall not be less than 29.6 volts at -10°F nor more than 32.5 volts at 120°F (See 4.2.4.4).

3.2.1.2.3 Heavy load. The Generator Head Assembly (A3174181) shall withstand repeated heavy loads of not less than 4 amperes for one minute with a cranking speed of 100 rpm (See 4.2.4.5).

3.2.1.2.4 Torque test. The Generator Head Assembly (A3174181) shall not require a torque greater than 80 inch-pounds when cranking at 70 rpm into a resistive load of 20 ohms. (See 4.2.4.6).

3.2.1.2.5 Protection. The Generator Head Assembly (A3174181) shall be capable of withstanding repeated momentary short circuits at one-minute intervals (See 4.2.4.7) and afterwards meet the requirements of 3.2.1.2.1.

3.2.1.2.6 Reverse current. The Generator Head Assembly (A3174181) shall not discharge the battery to which it is connected for charging; reverse current at 30 volts shall not exceed 750 microamperes with correct polarity and 50 milliamps with incorrect polarity. (See 4.2.4.8).

3.2.1.2.7 Acoustic Noise. The completely assembled G-67B/G (A3174128) shall be operated into a 20-ohm resistor and cranked by hand until 25 ±1 VDC are measured across the resistor. The sound level measured 5 feet away from the head shall be no greater than 70 dBA (See 4.2.4.20).

3.2.1.3 Environmental requirements. The Generator (A3174128) shall meet the following environmental requirements. (Generators (A3174128) shall be not be packaged unless otherwise specified).

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3.2.1.3.1 Temperature. Operability (3.2.1.5.6) and performance (3.2.1.2) requirements shall be met after storage at temperature extremes of -50°F and +160°F. Performance requirements (3.2.1.2.1 and 3.2.1.2.2) shall be met at operating temperatures of -50°F and 120°F (See 4.2.4.9 thru 4.2.4.10).

3.2.1.3.2 Vibration and shock. Operability (3.2.1.5.6) and performance (3.2.1.2.1 and 3.2.1.2.2) requirements shall be met after the generator (A3174128) undergoes vibration and shock (See 4.2.4.11 thru 4.2.4.15).

3.2.1.3.3 Humidity. Operability (3.2.1.5.6) and performance (3.2.1.2.1 and 3.2.1.2.2) requirements shall be met after prolonged exposure of the generator (A3174128) to humid conditions (See 4.2.4.16).

3.2.1.3.4 Immersion. Operability (3.2.1.5.6) and performance (3.2.1.2.1 and 3.2.1.2.2) requirements shall be met after the generator (A3174128) has been immersed in water. There shall be no leakage within the Generator Head Assembly (A3174181). (See 4.2.4.17)

3.2.1.3.5 Salt atmosphere. Operability (3.2.1.5.6) and performance (3.2.1.2.1 and 3.2.1.2.2) requirements shall be met after exposure of the unpackaged generator (A3174128) to salt atmospheres as may be encountered in coastal areas (See 4.2.4.18).

3.2.1.4 Reliability Requirements. The unit shall demonstrate a Mean Time Between Operational Mission Failure (MTBOMF) of no less than 790 operating hours when tested under the conditions specified. (See 4.2.4.19)

3.2.1.5 Construction. Construction shall be in accordance with A3174122 and the following.

3.2.1.5.1 Materials. Materials shall be those specified by the applicable drawings.

3.2.1.5.2 Accessories. Accessories shall be required as specified in the contract.

3.2.1.5.3 Interchangeability of Harmonic Drive. The harmonic drives used shall be interchangeable with one another, however, the components of any one harmonic drive shall not be interchanged with the components of another harmonic drive.

3.2.1.5.4 Protective finish. Unless specified on part drawing the following finish shall apply. All exposed aluminum surfaces shall be hard anodized per MIL-A-63576A, Type I except paragraphs 6.6 and 4.2.2.2. In paragraph 3.2.1.2 Type III, Class 2 coating shall be substituted for Type III, Class 1 coating as specified in MIL-A-8625. Final color shall be a lusterless black approximating FED-STD-595 final paint film designation AI. Exposed steel surfaces, such as screws, fasteners, and cables, shall be prepared per finish P211.1 of MIL-F-14072.

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Exposed stainless steel surfaces shall be passivated per Specification QQ-P-35. 300 Series stainless steel parts shall also be finished per MIL-C-13924 Class 4. All other external surfaces shall be made to approximate a lusterless black. Operating elements, such as spring loaded pins, and movable arms, and visual elements, such as meters and identification markings shall be usable after applying the finish.

3.2.1.5.5 Marking. The Generator Head Assembly (A3174181) shall be marked in accordance with MIL-M-13231. Panel marking shall be Group I as described in MIL-M-13231.

3.2.1.5.5.1 Serial numbers. The serial number of the Generator, Direct Current: G-67B/G, shall be assigned in accordance with MIL-M-13231.

3.2.1.5.5.2 Visibility. Identification markings on components and assemblies shall be so placed, wherever practical, so as to be readily visible in the final assembly.

3.2.1.5.6 Operability. The generator (A3174128) shall be capable of being set up in all operational and storage modes as per the drawing package. All sliding surfaces, rotating surfaces, and pins shall function without binding or restricted movement.

3.2.2 Physical Characteristics. The Generator, Direct Current, G67B/G shall be fabricated and assembled in accordance with Data List, DL A3174122.

3.3 Workmanship. The equipment shall be manufactured and assembled applying the applicable portion of the following paragraphs of MIL-P-11268.

a. General requirements for plastic material and plastic parts.

b. Wiring and cabling, including: slack, protection, clearance, splicing, connections in general, grounding, shielding on wire and cable.

c. Cleaning.

d. Riveting.

e. Welding.

Requirements of MIL-STD-454, Requirement 9, shall be used to establish acceptable workmanship criteria.

3.4 Samples.

3.4.1 First Article. When specified in the contract or purchase order, a sample of 20 generators shall be subjected to first article inspection (See 4.2.1.1 and 6.3.1).

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center frequency, MZ

In addition ten sets of the parts listed in section 6.4 shall be inspected per their applicable drawings.

3.4.2 Qualification Conformance Samples. As specified in the contract or purchase order samples of parts shall be subjected to inspection per 4.2.1.2.

3.4.3 Comparison Samples. As specified in the contract or purchase order one unit from each lot sample, refurbished as necessary, will be provided to the Government. These samples may be subjected to further testing to insure that first article requirements continue to be met. Failure of these samples will not affect acceptance of the lot, but may indicate further mandatory action regarding future quality conformance tests.

4. QUALITY ASSURANCE PROVISIONS

4.1 General.

4.1.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all required inspections as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspections specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1.1 Test equipment and inspection facilities. Test and measuring equipment and inspection facilities of sufficient accuracy, quality, and quantity to permit performance of the required inspection shall be established by the contractor and approved by the Government prior to use. Gage and instrument settings shall be a function of the equipment and independent of the operator. Test equipment shall be marked to indicate the limit of acceptance for each test.

4.1.1.2 Standard conditions. Tests shall be conducted under the following conditions unless otherwise specified:

Temperature	23° ± 10°C (73° ± 18°F)
Relative humidity	50 percent ± 30 percent
Atmospheric pressure	725 +50 -75 mm. Hg (28.5 +2.0 -3.0 in. Hg)

unless otherwise specified. The following tolerances on test equipment accuracy also apply, unless otherwise specified:

a.	Time	± 2 percent
b.	Resistance, current	± 2 percent

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4.1.1.3 Examination and tests. Each delivery shall be subjected to the examination and tests

c.	Frequency, period	± 2 percent
d.	Vibration amplitude	± 10 percent
e.	Force, weight, or torque	± 2 percent
f.	Voltage	± 0.5 percent
g.	Acoustic Noise Level	± 1.0 dBA

4.1.2 Special Tests and Examinations.

4.1.2.1 Performance of tests

4.1.2.1.1 Pretest performance records. Prior to proceeding with any of the environmental tests, the test item shall be operated under standard conditions (ref 4.2.1)to obtain data for determining satisfactory operation of the item as specified in the equipment specification, before, during and after the environmental test, as applicable. A record of specific pretest data shall be made to determine that the test item performs within prime item specification requirements. The pretest record shall include the functional parameters to be monitored during and after the test, if not specified in the equipment specification. This shall include acceptable functional limits (with permissible degradation) when operation of the test item is required.

4.1.2.1.2 Installation of test item in test facility. Unless otherwise specified, the test item shall be installed in the test facility in a manner that will simulate service usage, making connections and attaching instrumentation as necessary. Plugs, covers, and inspection plates not used in operation, but used in servicing shall remain in place. When mechanical or electrical connections are not used, the connections normally protected in service shall be adequately covered. For tests where temperature values are controlled, the test chamber shall be at standard ambient conditions when the test item is installed. The test item shall then be operated to determine that no malfunction or damage was caused due to faulty installation or handling. The requirement for operation following installation of the test item in the test facility is applicable only when operation is required during exposure to the specified test.

4.1.2.1.3 Performance check during test. When operation of the test item is required during the test exposure, suitable tests shall be performed to determine whether the test exposure is producing changes in performance when compared with pretest data.

4.1.2.1.4 Post-test data. At the completion of each environmental test, the test item shall be inspected in accordance with the equipment specification and the results shall be compared with the pretest data obtained in accordance with 4.1.2.1.1.

4.1.2.1.5 Test data. Test data shall include complete identification of all test equipment and accessories. The data shall include the actual test sequence used and ambient test conditions recorded periodically during the test period. The test record shall contain a signature and data block for certification of the test data by the test engineer.

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4.1.2.1.6 Failure criteria The item shall have failed the test when any of the following occur:

(1) Monitored functional parameters deviate beyond acceptable limits.

(2) Catastrophic electrical or structural failure.

(3) Mechanical binding or loose parts, including screws, clamps, bolts, and nuts, that clearly result in component failure or a hazard to personnel safety.

(4) Malfunction.

(5) Degradation of performance up to 5% beyond pretest record or equipment specification requirements (record to be made after test).

(6) Any additional deviations from acceptable criteria established before the test.

(7) Deterioration, corrosion, or change in tolerance limits of any internal or external parts which could in any manner prevent the test item from meeting operational service or maintenance requirements.

4.2 Quality Conformance Inspection.

4.2.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (does not include packaging) (See 4.2.1.1)

b. Quality conformance inspection of equipment before packaging. (See 4.2.1.2)

c. Quality conformance inspection of packaging (See 4.2.1.3)

d. Inspection covered by subsidiary documents (See 4.2.1.4).

4.2.1.1 First article inspection. The first article sample (FAS) shall consist of 20 units. These units shall be serialized 1 through 20 and evaluated in accordance with (1) the inspection protocol cited in Table II, and (2) the Classification of Defects cited in 4.2.2. In addition, (1) dimensional measurements certified by an independent laboratory approved by the Government will be provided for ten sets of each of the parts listed in 6.4 from the same parts lots used in building first-article-inspection-sample units and (2) Material and Component Certifications as described in 4.2.3 will be delivered.

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Table I Not Used

Table II. First Article Sample Inspection

Inspections	Req Para	Test Para	Serial Numbers To Be Used
Inspection Covered by Subsidiary Documents	-	4.5	1 - 20
All Group A tests	See Table III		1 - 20
Bounce preconditioning	3.2.1.1.2	4.2.4.2	1 - 20
Protection	3.2.1.2.5	4.2.4.7	1 - 20
Reverse Current	3.2.1.2.6	4.2.4.8	1 - 20
High Temperature	3.2.1.3.1	4.2.4.9	1, 2
Low Temperature	3.2.1.3.1	4.2.4.10	1, 2
Vibration	3.2.1.3.2	4.2.4.11	3, 4
Vibration-Temperature	3.2.1.3.2	4.2.4.12	4, 5
Shock (Drop)	3.2.1.3.2	4.2.4.13	5, 6
Shock (Drop-Temperature)	3.2.1.3.2	4.2.4.14	6, 7
Bounce	3.2.1.3.2	4.2.4.15	3, 7
Humidity	3.2.1.3.3	4.2.4.16	8
Immersion	3.2.1.3.4	4.2.4.17	9
Salt fog	3.2.1.3.5	4.2.4.18	10
Reliability	3.2.1.4	4.2.4.19	11 - 20

4.2.1.1.1 Reinstitution of tests. Acceptance of the First Article Sample (FAS) does not relieve the contractor from meeting all requirements of this specification throughout the contract. The Government reserves the right to independently verify all requirements by repeating FAS inspections or any portion thereof on any production lot. If such verification indicates failure to meet requirements previously verified by FAS tests, the contractor shall be required to take corrective action. If the requirements are not met on the first lot to which the correction can be applied, the Government reserves the right to institute tightened inspection on a lot basis, by adding to the normal quality conformance inspections all or such parts of the FAS tests needed to verify corrective action. Normal quality conformance inspection will be resumed after two consecutive lots have satisfactorily passed the tightened conditions (See 6.5).

4.2.1.2 Quality conformance inspection of equipment before packaging. Quality conformance inspection shall include inspection by subsidiary documents (4.2.1.4), classification of defects (4.2.2), material and component certifications (4.2.3), Group A Tests (Table III), and Group B Tests (Table IV). The contractor shall notify the cognizant Government QA representative prior to proceeding with inspection for Classification of Defects and Group A and B tests. Inspection may be made at a higher assembly level than

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specified provided that equal quality assurance provisions are satisfied but not at a lower level than specified. If a unit should fail any Group B test, testing shall be stopped and data and analysis shall be forwarded to the procuring agency within two working days. The procuring agency will evaluate the analysis and recommend changes where indicated.

4.2.1.2.1 Lot formation. A lot shall consist of units produced at one manufacturing location in accordance with stabilized methods and techniques, the same drawing, and specification, and revisions (except for those introduced without obsolescence), under continuous production. One lot shall comprise one month's production but not less than 50 units.

4.2.1.2.2 Sampling. Sampling shall be in accordance with MIL-STD-105 and this document. Unless otherwise specified, inspection level II shall apply.

4.2.1.2.3 Data recording. A record shall be made of the results of each performance test but only the defect summary need be recorded in the inspection of the classification of defects. The data shall be forwarded to the procuring agency within two working days of the Government acceptance of the item or lot. If inspections are conducted on a sublot basis, the acceptance data for each sublot shall be so identified.

4.2.1.2.4 Disposition of lot samples. After passing the prescribed tests, the samples shall be refurbished as necessary. One sample shall be selected as the comparison sample (See 6.5). The remaining samples shall be returned to the lot.

Table III. Group A Tests

Inspections	Requirement Paragraph	Test Paragraph	Acceptance Criteria
Burn in screening	3.2.1.1.1	4.2.4.1	100%
Rated output screening	3.2.1.2.1	4.2.4.3	100%
Output voltage screening	3.2.1.2.2	4.2.4.4	100%
Heavy load screening	3.2.1.2.3	4.2.4.5	100%
Torque screening	3.2.1.2.4	4.2.4.6	100%
Acoustic Noise screening	3.2.1.2.7	4.2.4.20	100%

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Table IV. Group B Tests

Inspections	Requirement Paragraph	Test Paragraph	Acceptance Criteria
Protection	3.2.1.2.5	4.2.4.7	AQL 1.0
Reverse current	3.2.1.2.6	4.2.4.8	AQL 1.0
High temperature	3.2.1.3.1	4.2.4.9	AQL 1.0
Low temperature	3.2.1.3.1	4.2.4.10	AQL 1.0

4.2.1.3 Quality conformance inspection of packaging. Packaging inspection shall be in accordance with the contract or higher level drawing.

4.2.1.4 Inspection covered by subsidiary documents. The following shall be inspected under the applicable subsidiary documents as part of the inspection required by this specification and the inspection requirements specified in the contract or purchase order.

Item	Where Required
Construction	3.2.1.5
Materials	3.2.1.5.1
Accessories	3.2.1.5.2
Finish	3.2.1.5.4
Marking	3.2.1.5.5
Workmanship	3.3
Preservation-Packaging	5.1

4.2.2 Classification of defects. The assembly shall be inspected by using MIL-STD-252 and the following Classification of Defects and applicable drawings.

4.2.2.1 Generator, Direct Current G-67B/G (A3174128)

Category	Defect	Method	Acceptance Criteria
Major			
101	Excessive Running torque	Appropriate Torque measuring instrument	100% Inspection
102	Excessive Acoustic Noise	DS-1	100% Inspection
Minor			
201	Incorrect Torque on all fasteners	Appropriate Torque measuring instrument	AQL 1.0
202	Incorrect Function of hinges	Manual	AQL 1.0
203	Excessive weld joints	Visual/DS-3	100% inspection

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204	Incorrect Finish	Visual	100% inspection

4.2.2.1.1 Power Module Assembly (A3174166)

Category	Defect	Method	Acceptance Criteria
Major			
101	Improper operation of slide knob	Visual	100% inspection
102	Improper operation of handles	Visual	100% inspection
103	Improper fit of crank shaft adapter	Visual	100% inspection
Minor			
201	Improper torque on all fasteners	Appropriate Torque measuring instrument	AQL 1.0

4.2.2.1.2 Generator Head Assembly (A31741811)

Category	Defect	Method	Acceptance Criteria
Major			
101	Lack of Water Seal	DS-2	2 samples/lot ac 0/re 1

4.2.2.1.3 Alternator/End Cap Assembly (A31741921).

Category	Defect	Method	Criteria
Minor			
201	Lack of adhesive on screws	Visual	AQL 1.0
202	Uncleanliness of harmonic drive	Visual	100% Inspection
203	Improper lubrication of harmonic drive	Visual	100% Inspection

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4.2.2.1.4 High Speed Shaft Assembly (A3174208)

Category	Defect	Method	Criteria
Major			
101	Damage to shaft or rotor	Visual (minimum magnification of 7X)	100% Inspection

4.2.2.1.5 Foot Support Assembly (A3174145)

Category	Defect	Method	Criteria
Major			
101	Improper fit of quick release pins	Visual	100% Inspection
102	Improper movement of hinged parts	Visual	100% Inspection
103	Incorrect weld on stirrup	Visual	100% Inspection
Minor			
201	Improper torque on all fasteners	Manual	AQL 1.0

4.2.2.1.6 Voltage Regulator Assembly (A3174232).

Category	Defect	Method	Acceptance Criteria
Major			
101	Improper parts Selection/ Placement	Visual	100% inspection
102	Improper soldering	Visual	100% inspection
103	Improper Conformal Coat	Visual	100% inspection
Minor			
201	Improper Torque on all Fasteners	Appropriate torque measuring instrument	100% inspection

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4.2.2.2 Defect standard.

DS-1 For the method of performing the acoustic noise test, refer to para 4.2.4.20

DS-2 Immerse in accordance the following procedure. The temperature of the water shall be 18° ±5°C (64° ±9°F) and the temperature of the test item shall be 45° ±3°C (113° ±5°F). The water container shall be of sufficient capacity so that the immersion of the test item will not raise the temperature of the water more than 3°C (5°F). Immerse the test item in the water so that the uppermost point of the test item is 36 ±5 inches below the surface of the water. The test item shall remain immersed for 120 ±5 minutes. Upon completion of the test period, remove the test item from the water and wipe the exterior surfaces of the test item dry. Open the test item and examine the interior and contents for evidence of leakage. Any evidence of moisture on disassembly shall constitute a failure.

DS-3 Individual part weldment drawings may specify additional test.

4.2.3 Material and Component Certification. Prior to acceptance under this procurement, the contractor shall demonstrate by means of certifications or statements of findings that only materials and components conforming to the item specification, drawings, and subsidiary documents (4.2.1.4) have been used.

4.2.4 Test methods.

4.2.4.1 Burn-In. The Generator Head Assembly (A3174181) shall be connected to an electric motor and operated over a 1 1/2 hour period (1/2 hour on, 1/2 hour off, 1/2 hour on) at an input rotational speed of 68 rpm with the output terminals connected to a 20-ohm load.

4.2.4.2 Bounce preconditioning. The Generator Head Assembly (A3174181) shall be mounted on a suitable fixture and then placed on the table of a package tester (type 1000 SC, L.A.B. Corporation, Skaneatales, NY, or equivalent). The package tester, shafts in phase, shall have a speed such that the items bounce 1/4 inch from the bed in a continuous cycle. The items shall be subjected to this preconditioning for 10 minutes. After the bounce preconditioning, the unit shall not be repaired, aligned, cleaned or otherwise changed prior to being subjected to performance requirements. The generator head shall meet the requirements of 3.2.1.1.2.

4.2.4.3 Rated output. The Generator Head Assembly (A3174181) shall be connected to a resistive load of 10.0 ohms and driven at 85 rpm. The output current shall be measured with an external ammeter and shall meet the requirements of 3.2.1.2.1.

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4.2.4.4 Output voltage. The Generator Head Assembly (A3174181) shall be driven at 105 rpm into a 100-ohm load. The output voltage shall be measured with an external voltmeter and shall meet the requirements of 3.2.1.2.2

4.2.4.5 Heavy load. The Generator Head Assembly (A3151481) shall be driven at 100 rpm, and the resistance load shall be adjusted to obtain an output current of 4 (+0.25, -0.00) amperes. The Generator Head Assembly (A3174181) shall be so driven 10 times for 1.0 minute periods with 5-minute intervals between tests during which time the Generator Head Assembly (A3174181) shall not be cranked at all. The Generator Head Assembly (A3174181) shall meet the rated output requirements 3.2.1.2.1 after this test.

4.2.4.6 Torque. Connect a 20-ohm load across the output connectors. Run the Generator Head Assembly (A3174181) at 70 rpm for 5 minutes before taking the measurement. The unit shall meet the torque requirements of 3.2.1.2.4.

4.2.4.7 Protection. The Generator Head Assembly (A3174181) shall be driven at 75 rpm, and a series of short circuits of between one and two seconds duration shall be applied across the output terminals. The output shall be short circuited a minimum of five times with one-minute intervals between shorting. The cranking speed shall be maintained during each short circuit. After the test, the Generator Head Assembly (A3174181) shall meet the requirements of 3.2.1.2.5.

4.2.4.8 Reverse current. A DC power supply with an adjustable output voltage from 0 to 30 VDC shall be connected to the output terminals of the Generator Head Assembly (A3174181) in correct polarity, with the Generator Head Assembly (A3174181) not being cranked. The voltage of the power supply shall be gradually increased to 30 VDC and the reverse current shall not exceed the value specified in 3.2.1.2.6. This test shall be repeated with the power supply connected incorrectly to the output terminals of the generator head. i.e. positive to negative. The current shall not exceed the value specified in 3.2.1.2.6. The Generator Head Assembly (A3174181) shall meet the rated output requirement 3.2.1.2.1 after this test.

4.2.4.9 High temperature. The Generator Head Assembly (A3174181) shall be temperature conditioned in accordance with the following procedure:

Step 1 - Perform the procedures defined in 4.1.2.1.

Step 2 - Raise the internal chamber temperature to 49°C (120°F).

Step 3 - Maintain internal chamber temperature for 6 hours at 49°C (120°F).

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- Due to the damping of the meter, the pointer will increase and

Step 4 - Raise the internal chamber temperature to 71°C (160°F) within a time period of 1 hour and then maintain at that temperature for 4 additional hours.

Step 5 - Lower the internal chamber temperature to 49°C (120°F) within a time period of 1 hour.

Step 6 - Repeat steps 3, 4, and 5 two additional times (making a total of three 12-hour cycles).

Step 7 - Return the test item, nonoperative, to standard ambient conditions and stabilize.

At least once during the three 12 hour cycles after the unit has stabilized at 120°F, once at 120°F after the third cycle, and once at ambient temperature upon completion of the test, the Generator Head Assembly (A3174181) shall undergo the rated output test (4.2.4.3) and the output voltage test (4.2.4.4) and the applicable requirements of paragraphs 3.2.1.2.1 and 3.2.1.2.2.

4.2.4.10 Low temperature. The Generator Head Assembly (A3174181) shall be temperature conditioned in accordance with the following procedure:

Step 1 - Perform the procedures defined in 4.1.2.1.

Step 2 - Lower the internal chamber temperature to the storage temperature (-50°F) maintain for a period of 24 hours after stabilization.

Step 3 - Return the test item, non-operating, to standard ambient conditions and stabilize.

NOTE: The rate of temperature change may be the maximum attainable by the chamber but shall not exceed 10°C (18°F) per minute.

At least once during the -50°F portion of the test after the unit has stabilized and once at ambient temperature upon completion of the test, the Generator Head Assembly (A3174181) shall undergo the rated output test (4.2.4.3) and the output voltage test (4.2.4.4) and satisfy the applicable requirements of paragraphs 3.2.1.2.1 and 3.2.1.2.2.

4.2.4.11 Vibration test. Secure the unpackaged generator (A3174128) under test to an appropriate vibrator table. The test item shall be vibrated along each axis in accordance with **figure 1**. The frequency of applied vibration shall be swept over the specified range logarithmically in accordance with **figure 2**. The vibrating frequency shall be swept logarithmically for 84 minutes per axis with a sweep time of 12 minutes per 5-200-5 Hz cycle. The Generator Head Assembly(A3174181) shall be tested and shall then meet the requirements of 3.2.1.3.2.

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4.2.4.12 Vibration-temperature test. The generator (A3174128) in its combat carrying bag (A3174126) with the cable (A3174124) in the bag pouch shall be tied down to the vibration fixture in accordance with the normally prescribed tie- down procedures so that vibration is applied through axis of the generator. The Generator Head Assembly (A3174181) shall be at a temperature of 73°F. The generator shall be vibrated for two hours in accordance with the random vibration spectrum shown in **figure 3**. The inputs specified are those associated with the fixture and not the generator. The control accelerometer shall be on the fixture. Input tolerances for random vibration are to be held to ± 3 dB up to 200 Hz and ± 5 dB over 200 Hz.

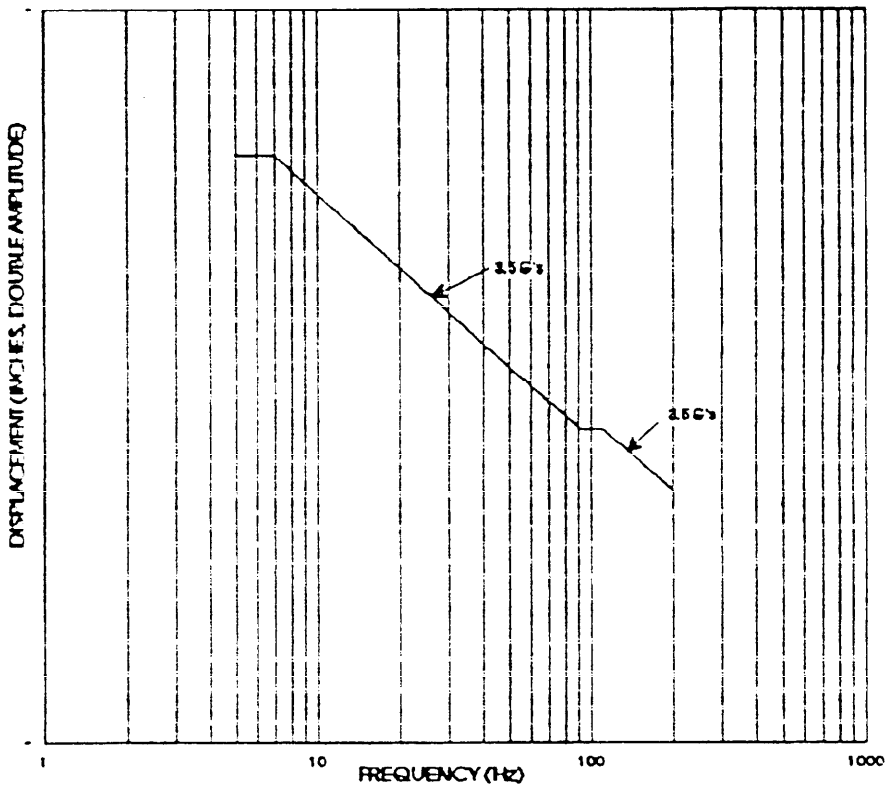


figure 1

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At completion, the Generator Head Assembly (A3174181) shall be subjected to the following sinusoidal vibration environment:

Frequency	Vibration Amplitude	Sweep Rate	Total Time/Axis
(Hz)	(g)		(min)
10 - 50	1.5	28 Hz/min	5
50 - 150	2.5	28 Hz/min	

The entire procedure described above shall be repeated two more times for a total of three cycles, first with the vibration imparted through the generator (A3174128) longitudinal axis, and second with the vibration imparted through the generator (A3174128) transverse axis. At the completion of each of the three cycles, the generator (A3174128) shall meet the requirements of paragraphs 3.2.1.3.2 and 3.2.1.2.1. The entire procedure described above shall be repeated two more times, once at a temperature of

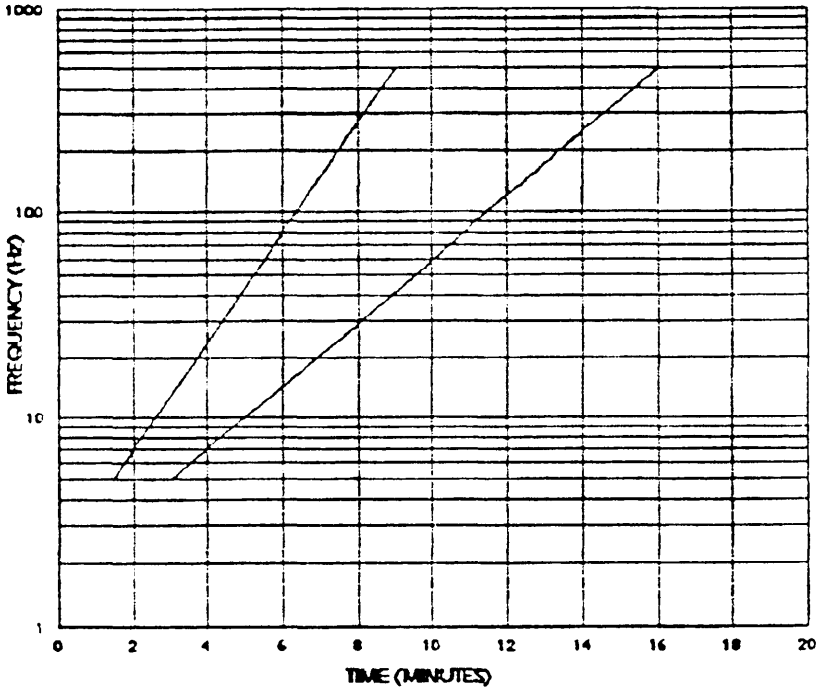


figure 2

+130° ± 2.5°F and again at a temperature of -30° ± 2.5°F. These temperature extremes are to be achieved at a rate of 25 degrees per hour, and the generator (A3174128) shall be stabilized at the test temperature for at least 16 hours before vibration. The control temperature sensor shall be attached to the outer surface of the generator (A3174128) under test. During changes in

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vibration axes, the generator (A3174128) test temperature shall be maintained. The generator (A3174128) shall be returned to ambient temperature before the functional tests and appropriate inspections are conducted. After the completion of the high temperature testing, the generator (A3174128) shall meet the requirements of 3.2.1.3.2.

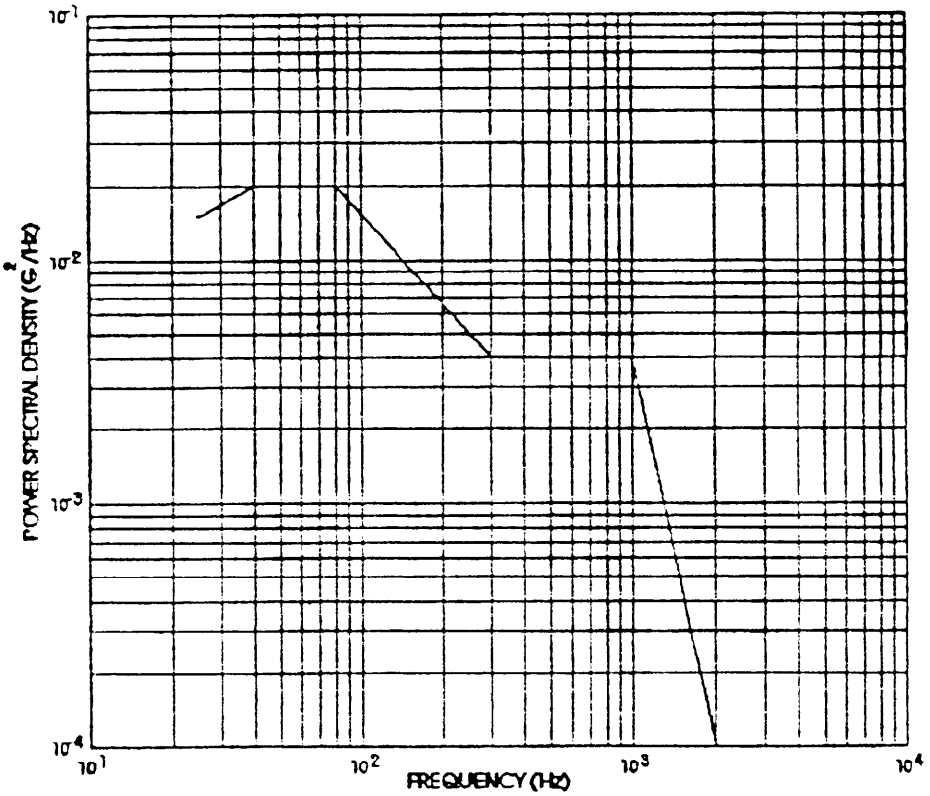


figure 3

4.2.4.13 Shock (Transit Drop) test. The generator (A3174128) shall be packed in its carrying bag (A3174126) with the cable (A3174124) in the bag pouch. The floor or barrier receiving the impact shall be of 2-inch plywood backed by concrete. The generator shall be dropped from a height of 48-inches. The generator shall be dropped on each face, and corner for a total of 14 drops. Drops shall be made from a quick-release hook, or drop tester. The test item shall be so oriented that upon impact a line from the struck corner or edge to the center of gravity of the case and contents is perpendicular to the impact surface. Upon completion of the test, the generator (A3174128) shall be capable of operation by hand and shall meet the requirements of 3.2.1.3.2 except 3.2.1.5.6.

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4.2.4.14 Shock (Drop-Temperature) test. The generator (A3174128) in the carrying bag (A3174126) with the cable (A3174124) in the bag pouch shall be placed inside a conditioning chamber at ambient conditions. The chamber temperature shall be increased at the rate of 25 degrees per hour and stabilized at +130° ± 2.5°F for a period of not less than 16 hours. The generator (A3174128) in the carrying bag shall be removed from the conditioning chamber and raised to a height such that its lowest corner is 30 inches above a concrete impact surface. The Generator Head Assembly (A3174181) shall be the impacting part of the generator (A3174128). Within five minutes after removal from the conditioning, the generator (A3174128) shall be released using a quick-release mechanism. The generator (A3174128) shall be oriented at release such that its center of gravity is directly above the striking corner. After visually inspecting the generator (A3174128) and carrying bag (A3174126) for damage, it shall be returned to the conditioning chamber and reconditioned at +130° ± 2.5°F for a period of not less than 1 hour and dropped on a different corner and inspected. This procedure will be followed until the generator (A3174128) has been dropped on all four of the generator's corners. At the completion of the four drops and inspections, the generator (A3174128) shall meet the requirements of 3.2.1.3.2. Using a second generator (A3174128) and combat carrying bag, the entire procedure described above shall be repeated but at a conditioning temperature of -30° ± 2.5°F.

4.2.4.15 Bounce (loose cargo) test. The packaged generator (A3174128) shall be tested in accordance with the following procedure:

Apparatus. A package tester capable of 1-inch (double amplitude) displacement and of suitable capacity for testing military equipment.

Test conditions. The test bed of the package tester shall be covered with a panel of 1/2-inch plywood, with the grain parallel to the drive chain. The plywood shall be secured with six-penny nails spaced at 6-inch intervals around all four edges. If the distance between either pair of fences is greater than 24 inches, the plywood shall also be nailed at 3-inch intervals in a 6-inch square at the center of the test area. Using suitable wooden fences, constrain the test item to a horizontal motion of not more than 2 inches in a direction parallel to the axes of the axes of the shafts, a distance more than sufficient to insure the test item will not rebound from fence to fence.

The generator (A3174128) in its carrying case (A3174126) with the cable (A3174124) in the bag pouch shall be placed on the package tester within the constraints outlined above. The test item will not be operated during vibration. The package tester shall be operated in the synchronous mode with the shafts in phase. (In this mode any point on the bed of the package tester will move in a circular path in a vertical plane perpendicular to the axes of the shafts.) The package tester shall be operated at 1-inch double amplitude and 284 rpm +2 rpm for a total of 3 hours. At

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the end of each 1/2-hour period, turn the test item to rest on a different face, so that at the end of the 3-hour period the test item will have rested on each of its six faces (top, bottom, sides, and ends). At the end of the 3-hour period, the test item shall be operated and inspected and shall meet requirements of 3.2.1.3.2

4.2.4.16 Humidity test. The generator (A3174128) shall be tested in accordance with the following procedure:

Step 1 - Perform the procedure defined in 4.1.2.1.

Step 2 - Dry the test item at 54°C (129°F) for 24 hours.

Step 3 - Condition the test item at 23°C (73°F) and 50 ±10 percent relative humidity for 24 hours.

Step 4 - Raise the internal chamber temperature to 30°C (86°F) and the relative humidity to 94 ±4 percent.

Step 5 - Subject the test item to five continuous 48-hour cycles in accordance with **figure 4**. The relative humidity shall be maintained at 94 ±4 percent at all times.

Step 6 - The generator (A3174128) shall meet the requirements of 3.2.1.3.3 after conditioning.

NOTE: The chamber and accessories shall be constructed and arranged in such a manner as to avoid condensate dripping on the test item. The chamber shall be trap-vented to the atmosphere to prevent the buildup of total pressure. Relative humidity shall be determined from the dry bulb-wet bulb thermometer comparison method or an equivalent method approved by the procuring activity. When readout charts are used, they shall be capable of being read with a resolution within 0.6°C (1°F). When the wet bulb control method is used, the wet bulb and tank shall be cleaned and a new wick installed at least every 30 days. The air velocity flowing across the wet bulb shall be not less than 900 feet per minute. Provisions shall be made for controlling the flow of air throughout the internal chamber test space where the velocity of air shall not exceed 150 feet per minute. Steam or distilled, demineralized, or deionized water having a pH value between 6.0

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and 7.2 at 23°C (73°F) shall be used to obtain the specified humidity. No rust or corrosive contaminants shall be imposed on the test item by the test facility. The test item may be realigned or readjusted as necessary following Step 2 to meet specification requirements. No further realignment or readjustment shall be permitted throughout the test period other than with accessible controls employed for operation of the test item. No repair or replacement of parts shall be permitted.

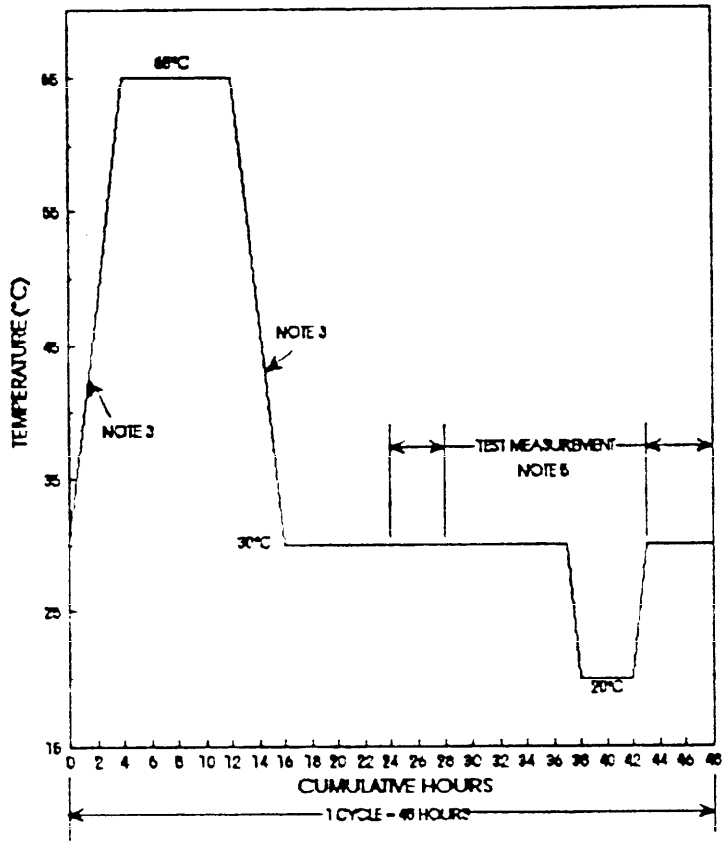


figure 4

NOTES:

1. Tolerance during temperature change shall not be greater than 3°C (5°F).

2. Relative humidity shall be maintained at 94 ±4 percent at all times, except that during the descending temperature period, the relative humidity may be permitted to drop as low as 85 percent.

3. Rate of temperature change between 30° and 65°C (86° and 149°F) shall be not less than 8°C (14.4°F) per hour.

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4. Test measurements shall be taken only at the period specified in the applicable equipment or system specification.

4.2.4.17 Immersion test. The generator (A3174128) shall be tested in accordance with the following procedure:

A container of adequate size to hold the generator (A3174128) shall be filled with water. The temperature of the water shall be 18° ±5°C (64° ±9°F) and the temperature of the test item shall be 45° ±3°C (113° ±5°F). The water container shall be of sufficient capacity so that the immersion of the test item will not raise the temperature of the water more than 3°C (5°F). The generator (A3174128) shall be immersed in the water so that the uppermost point of the test item is 36 ±5 inches below the surface of the water. The test item shall remain immersed for 120 ±5 minutes. Upon completion of the test period, remove the test item from the water and wipe the exterior surfaces of the test item dry. Open the test item and examine the interior and contents for evidence of leakage. The generator (A3174128) shall meet the requirements of 3.2.1.3.4.

4.2.4.18 Salt fog test. The generator (A3174128) shall be tested in accordance with the following procedure:

The apparatus used in the salt fog test shall include the following:

a. Exposure chamber with racks for supporting test items.

b. Salt solution reservoir with means for maintaining an adequate level of solution.

c. Means for atomizing salt solution, including suitable nozzles and compressed air supply.

d. Chamber heating means and control.

e. Means for humidifying the air at a temperature above the chamber temperature.

The chamber and all accessories shall be made of material that will not contaminate and affect the corrosiveness of the fog, e.g., glass, hard rubber, plastic, or kiln dried wood other than plywood. In addition, all parts which come in contact with test items shall be of materials that will not cause electrolytic corrosion. The chamber and accessories shall be constructed and arranged so that there is no direct impingement of the fog or dripping of the condensate on the test items, that the fog circulates freely about all test items to the same degree, and that no liquid which has come in contact with the test items returns to the salt-solution reservoir. The chamber shall be

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properly vented to prevent pressure buildup and allow uniform distribution of salt fog. The discharge end of the vent shall be protected from strong drafts which can create strong air currents in the test chamber. The atomizers used shall be of such design and construction as to produce a finely divided, wet, dense fog. Atomizing nozzles shall be made of material that is nonreactive to the salt solution. The compressed air entering the atomizer shall be essentially free from all impurities, such as oil and dirt. Means shall be provided to humidify and warm the compressed air as required to meet the operating conditions. The air pressure shall be suitable to produce a finely divided dense fog with the atomizer or atomizers used. To insure against clogging the atomizers by salt deposition, the air should have a relative humidity of at least 85 percent at the point of release from the nozzle. A satisfactory method is to pass the air in very fine bubbles through a tower containing heated water which should be automatically maintained at a constant level. The temperature of the water should be at least 35°C (95°F). The permissible water temperature increases with increasing volume of air and with decreasing insulation of the chamber and the chamber's surroundings. However, the temperature should not exceed a value above which an access of moisture is introduced into the chamber (for example 43°C (109°F) at an air pressure of 12 psi) or a value which makes it impossible to meet the requirements for operating temperature.

The salt used shall be sodium chloride containing on the dry basis not more than 0.1 percent sodium iodide and not more than 0.5 percent of total impurities. A 5 ±1 percent solution shall be prepared by dissolving five parts by weight of salt in 95 parts by weight of distilled or demineralized water. The solution shall be adjusted to and maintained at a specific gravity between the limits shown on **figure** 5 by utilizing the measured temperature and density of the salt solution. Sodium tetraborate (common borax) may be added to the salt solution in a ratio not to exceed 0.7 grams (1/4 level teaspoon) sodium tetraborate to 20 gallons of slat solution as a pH stabilization agent. Adjustment of pH of the salt solution shall be so maintained that the solution atomized at 35°C (95°F) and collected by the method specified in 3.1.3 will be in the pH range of 6.5 to 7.2. Only diluted C.P. hydrochloric acid or C.P. sodium hydroxide shall be used to adjust the pH. The addition of sodium tetraborate as recommended above will aid in maintaining a stable pH value. The pH measurement shall be made electrometrically, using a glass electrode with a saturated potassium chloride bridge, or by a colorimetric method, such as bromothymol blue, provided the results are equivalent to those obtained with the electrometric method. The pH shall be measured when preparing each new batch of solution

The test shall be conducted with a temperature in the exposure zone maintained at 35°C (95°F) The use of immersion

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heaters within the chamber for the purpose of maintaining the temperature within the exposure zone is prohibited.

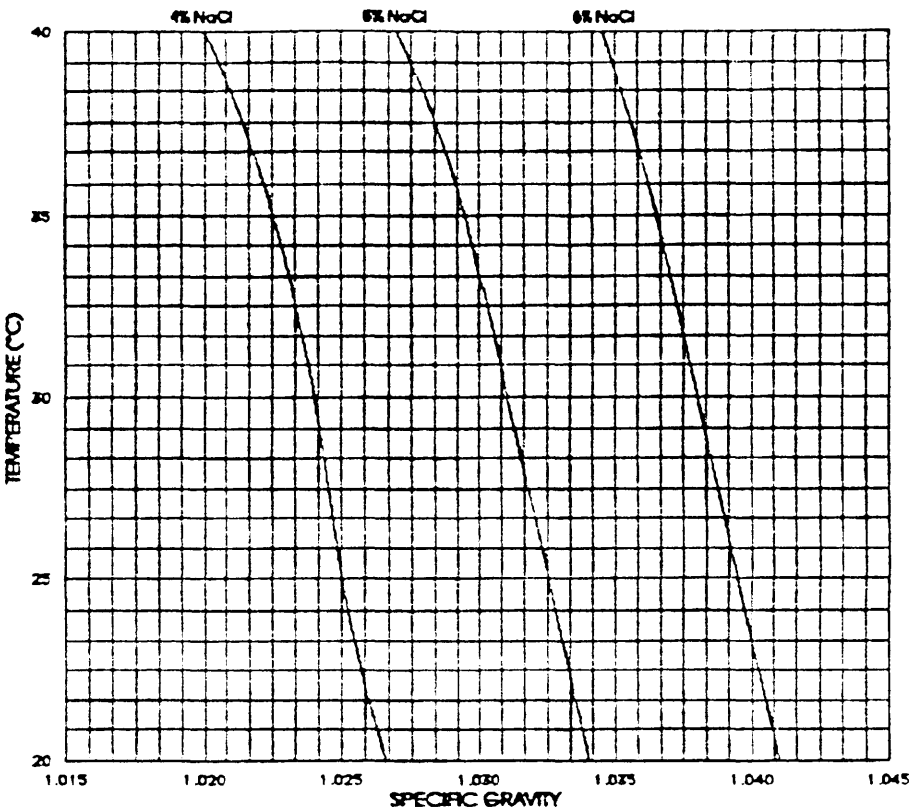


figure 5

The salt fog conditions maintained in all parts of the exposure zone shall be such that a clean fog collecting receptacle placed at any point in the exposure zone will collect from 0.5 to 3 milliliters of solution per hour for each 80 square centimeters of horizontal collecting area (10 centimeters diameter) based on an average test of at least 16 hours. A minimum of two receptacles shall be used, one placed nearest to any nozzle and one farthest from all nozzles. Receptacles shall be placed so that they are not shielded by test items and so not drops of solution from test items or other sources will be collected. The solution, collected shall have the sodium chloride content and pH specified above when measured at a temperature of 35°C (95°F). The salt solution from all collection receptacles used can be combined to provide that quantity required for the measurements specified.

The measurement of both sodium chloride content and pH shall be made following each test.

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The test item shall be given a minimum of handling, particularly on the significant surfaces, and shall be prepared for test immediately before exposure Uncoated metallic or metallic coated devices shall be thoroughly cleaned of oil, dirt, and grease as necessary until the surface is free from water break. The cleaning methods shall not include the use of corrosive solvents nor solvents which deposit either corrosive or protective films, nor the use of abrasives other than a paste of pure magnesium oxide.

The test item shall be placed in the test chamber and exposed to the salt fog for a period of 48 hours. After conditioning, visual inspection shall show no flaked, pitted, blistered, or otherwise loosened finish on metal surface. If necessary to aid in examination, a gentle wash in running water not warmer than 38°C (100°F) may be used. The test item shall then be stored in an ambient atmosphere for 18 hours. The generator (A3174128) shall meet the requirements of 3.2.1.3.5.

4.2.4.19 Reliability test. The reliability test shall demonstrate the MTBOMF required in paragraph 3.2.1.4. Testing shall be conducted in accordance with MIL-HDBK-781 Fixed Duration Test Plan XXI-D. The Generator Head Assembly (A3174181) shall be connected to a resistive load of 15.0 ohms and cranked at a speed of 68 rpm and shall be operated in a standard ambient environment for a period of 1 hour. The Generator Head Assembly (A3174181) shall then be allowed to rest at ambient conditions for 1 hour. This cycle shall be repeated until each generator head (A3174181) has accumulated a total of 87 operating hours each.

The requirements of Probability Ratio Sequential Test Plan VI-D of MIL-HDBK-781 may be substituted for Test Plan XXI-D in the paragraph above.

Voltage output shall be continuously monitored. If the voltage of any Generator Head Assembly (A3174181) drops below 20 volts at any time during the test this unit shall be scored as a failure.

4.2.4.20 Acoustic noise. A sound meter, GEN RAD 1982 Precision Sound Meter and Analyzer, or equivalent, shall be placed so that the tip of the microphone is directly in front of the generator head (A3174128), on a level with it and at a distance of five feet from it. The long axis of the meter shall be at an angle

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of 20 degrees to the line joining the generator head and the tip of the microphone. The d-c voltage output shall be connected to a 20-ohm resistor and the voltage read with an appropriate meter. The generator (A3174128) shall be operated by hand in a location having a background noise level of 45 dBA or less. Sound level readings shall be made when the voltage output under the conditions described is 25 ± 1 volts. Acoustic measurements shall be made on the dBA scale of the sound level meter and shall meet the requirements of 3.2.1.2.7.

5. PREPARATION FOR DELIVERY

5.1 Preservation-packing. Preservation and packing shall be in accordance with the contract or higher level documentation.

5.1.1 Shipping containers. Shipping containers shall be as specified in the contract or higher level documentation.

5.1.2 Marking. Marking shall be as specified in the contract or higher level documentation.

5.1.3 Military. In addition to any special marking required by the contract or order, interior packs and exterior shipping containers shall be marked in accordance with 3.2.1.5.5 and MIL-STD-129.

6. NOTES

6.1 Intended use. The intended use of the G-67B/G hand-cranked generator is to provide power to charge batteries in forward areas, and to operate radios if required.

6.2 Ordering information

6.2.1 Documents. Procurement documents should specify the title, date, and number of this specification.

6.2.2 Data Requirements. When this specification is used in an acquisition and data are required to be delivered, the data requirements identified below shall be developed as specified by

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an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved Contract Data Requirements List (CDRL), incorporated into the contract. When the provisions of DOD FAR supplement, Part 27, Subpart 2.7.410-6 (DD Form 1423) are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification are cited in the following paragraphs:

Para. No.	Data Requirement Title	Appl DID No.
4.2.1.1	First Article Qualification Test Plan	DI-T-5315A
4.2.1.2	Lot Summary Inspection Record	DI-MISC-81016
4.1.1.1	Special Inspection Equipment Concepts	DI-QCIC-80757
4.2.1.2	Failure Analysis and Corrective Action Report	DI-R-5299C

6.2.3 Engineering information. Upon request, copies of the data collected in the acceptance of a lot will be made available to the Government representative. Additional data or data involving special requirements, when needed for engineering purposes, will be specifically defined by the contract.

6.3 Instructions for Model and Samples.

6.3.1 First Article. When a first article inspection is required, the items inspected shall be a first article sample. The first article sample should consist of twenty units. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and dispositions of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.3.1.1 Lead time. Government approval of the first article will require a maximum of twenty working days if there are no reported failures. If there are failures, additional time will be required.

6.3.1.2 Disposition of First Article Samples. After inspection, the first article sample may be submitted with a subsequent lot after being washed, cleaned, dried, and refurbished as necessary. The units will be subjected to quality conformance tests for that lot.

6.4 First article sample parts list. Ten sets of the following parts shall be compared with drawings as specified.

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Drawing No	Title	Characteristic
A3174124	CABLE, ELECTRICAL & CONNECTING PLUG	ALL PHYSICAL
A3174124-1	BACK SHELL ADAPTER	ALL PHYSICAL
A3174125-2	BACK SHELL ADAPTER	ALL PHYSICAL
A3174126	BAG, CARRYING	ALL PHYSICAL
A3174127	STRAP ASSEMBLY	ALL PHYSICAL
A3174132	STOP, SLEEVE	ALL PHYSICAL
A3174133	LANYARD	ALL PHYSICAL
A3174134-1	PIN, PULL, 1.25	ALL PHYSICAL
A3174134-2	PIN, PULL, 1.50	ALL PHYSICAL
A3174135	STAND	ALL PHYSICAL
A3174137	STAND COLLAR	ALL PHYSICAL
A3174138	STAND LEG	ALL PHYSICAL
A3174140	BRACKET & OUTER TUBE ASSEMBLY	ALL PHYSICAL
A3174142	BRACKET, FOOT SUPPORT	ALL PHYSICAL
A3174143	OUTER TUBE, FOOT SUPPORT	ALL PHYSICAL
A3174144	STOP, OUTER TUBE	ALL PHYSICAL
A3174147-1	HINGE PINS, 1.00" LENGTH	ALL PHYSICAL
A3174147-2	HINGE PINS, 1.25" LENGTH	ALL PHYSICAL
A3174150	UPPER STIRRUP SLEEVE	ALL PHYSICAL
A3174151	STIRRUP	ALL PHYSICAL
A3174152	LOWER STIRRUP SLEEVE	ALL PHYSICAL
A3174156	THUMBWHEEL	ALL PHYSICAL
A3174157	FOOTPAD	ALL PHYSICAL
A3174161	INNER TUBE & HINGE ASSEMBLY	ALL PHYSICAL
A3174163	INNER TUBE, FOOT SUPPORT	ALL PHYSICAL
A3174164	HINGE	ALL PHYSICAL
A3174167	NUT, HANDLE	ALL PHYSICAL
A3174172	SHAFT, HANDLE	ALL PHYSICAL
A3174173	GRIP, HANDLE	ALL PHYSICAL
A3174176	SLIDE KNOB	ALL PHYSICAL
A3174177	PLUNGER	ALL PHYSICAL
A3174178	CRANK ARM, MACHINED	ALL PHYSICAL
A3174179	CRANK SHAFT ADAPTER	ALL PHYSICAL
A3174180	DUST COVER	ALL PHYSICAL
A3174182	PLATE, IDENTIFICATION, G-67B/G	ALL PHYSICAL
A3174186	SEAL RETAINER	ALL PHYSICAL
A3174187	SEAL, SHAFT	ALL PHYSICAL
A3174194	BEARING SPACER	ALL PHYSICAL
A3174196	ACOUSTIC END CAP, POTTED	ALL PHYSICAL
A3174202	ACOUSTIC END CAP WEB	ALL PHYSICAL
A3174203	ACOUSTIC END CAP	ALL PHYSICAL
A3174208	HIGH SPEED SHAFT ASSEMBLY	ALL PHYSICAL
A3174210	STOP, ROTOR	ALL PHYSICAL

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Drawing No	Title	Characteristic
A3174211	ROTOR, ALTERNATOR	ALL PHYSICAL
A3174213	HIGH SPEED SHAFT	ALL PHYSICAL
A3174214	ALTERNATOR HOUSING ASSEMBLY	ALL PHYSICAL
A3174216	STATOR, ALTERNATOR	ALL PHYSICAL
A3174218	HOUSING	ALL PHYSICAL
A3174223	LOW SPEED SHAFT	ALL PHYSICAL
A3174225	CASE	ALL PHYSICAL
A3174226	METER ASSEMBLY	ALL PHYSICAL
A3174230	BOTTOM, CASE	ALL PHYSICAL
A3174232	VOLTAGE REGULATOR MODULE	ALL PHYSICAL
A3174234	HEAT SINK	ALL PHYSICAL
A3174235	STANDOFF	ALL PHYSICAL
A3174236	TRANSISTOR, DARLINGTON, NPN	ALL PHYSICAL
A3174237	TRANSISTOR, MOUNT	ALL PHYSICAL
A3174240	PRINTED CIRCUIT BOARD ASSEMBLY	ALL PHYSICAL
A3174243	PRINTED CIRCUIT BOARD	ALL PHYSICAL
A3174244-1	PIN AND RECEPTACLE	ALL PHYSICAL
A3174244-2	PIN AND RECEPTACLE	ALL PHYSICAL
A3174245	RESISTOR, 6.3 KΩ, 3 W	ALL PHYSICAL

6.5 Comparison Sample. The contract should provide for delivery to a Government technical laboratory (if such a laboratory is defined in the contract) of one unit from each lot sample, refurbished as necessary, within two days of lot acceptance for use in comparison sample tests (See 3.4.3). These samples are subject to periodic retesting by the Government to insure that first article requirements continue to be met. Failure of these samples will not affect acceptance of the lot, but may indicate future mandatory action regarding testing of future quality conformance lots.

6.6 Fungus, Blowing sand and dust, and Electromagnetic compatibility. These tests have been succesfully performed in the past by Harry Diamond Laboratories on similiar generators. They are not necessary unless the procuring activity believes that designed changes have been made that require these issues to be readdressed. Specific tests should be generated at the time of procurement to meet current environmental requirements.

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Custodians:
Army - HD

Preparing Activity:
Army - HD

(Project 6115-A586)

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